

Financial Complaint Auto-Routing with NLP

Champion-challenger NLP decision-support prototype

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Executive summary. This internal prototype compares leakage-safe TF-IDF + Linear SVM with a frozen DistilBERT challenger and development-only routing policies across eight CFPB product categories.

Scope. 2024: 50,000 raw rows -> 33,042 corrected rows -> 26,433 development / 6,609 internal test.
2025: 30,156-row primary headline cohort / 49,225-row secondary sensitivity cohort; retrospective, not a new untouched V2 holdout.

Business Problem

Complaint operations teams must route unstructured narratives to the correct product area. Manual triage can be slow and inconsistent, while uncontrolled automation can misroute complex or higher-risk cases. The goal is to support consistent initial recommendations while preserving human review for weaker or ambiguous model signals.

Decision-Support Workflow



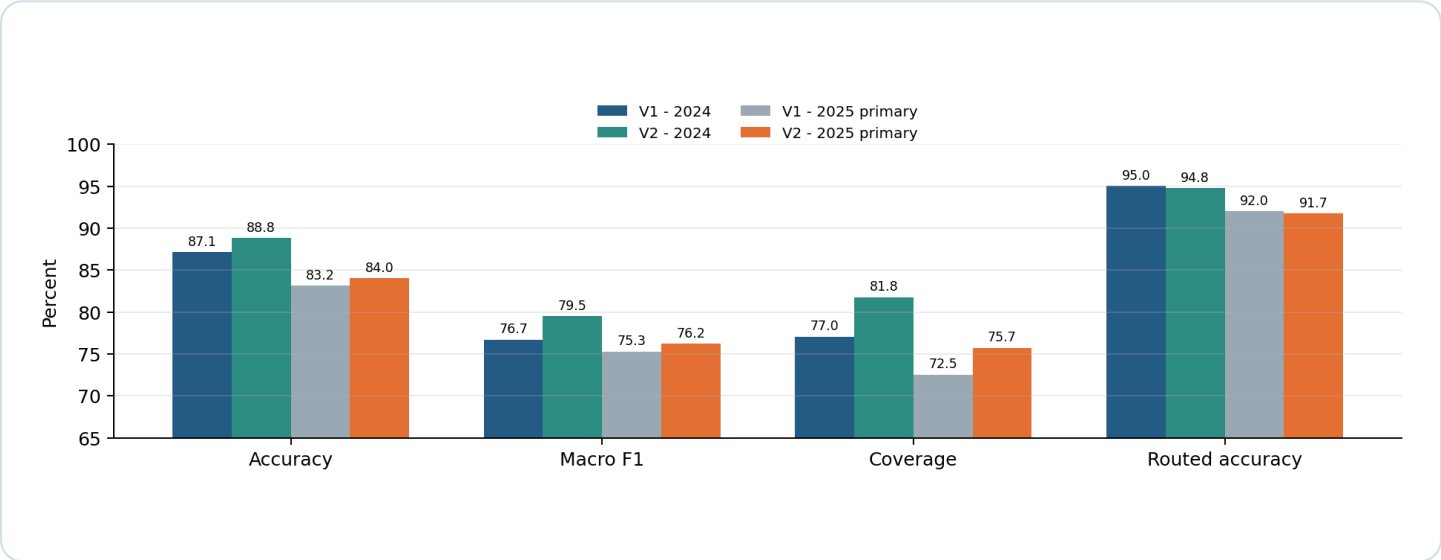
FINAL OUTCOME Retain Version 1 as the temporally validated benchmark. Version 2 improved shared-benchmark Macro F1 and routing coverage, but it did not establish a clear routed-risk or operational advantage. It remains a frozen challenger pending evaluation on a new untouched period.

Shared 2024 Benchmark: V1 vs. V2

Weighted F1: V1 0.8715 | V2 0.8859 (+0.0144)

ACCURACY		MACRO F1		COVERAGE		ROUTED ACCURACY		MISROUTE RATE	
V1	87.12%	V1	76.71%	V1	77.05%	V1	95.03%	V1	4.97%
V2	88.82%	V2	79.49%	V2	81.77%	V2	94.76%	V2	5.24%
+1.69 pts		+2.78 pts		+4.72 pts		-0.27 pts		+0.27 pts	

2024 Shared Benchmark and Retrospective 2025 Primary Cohort



Headline finding. Both models weakened on the stricter 30,156-row 2025 primary cohort. V2 retained small classification and coverage gains over V1, but its primary routed accuracy was slightly lower (91.74% vs. 92.03%) and its misroute rate slightly higher (8.26% vs. 7.97%).

Why the Evaluation Is Credible

Leakage-safe evaluation, frozen policies, and temporal evidence

TECHNICAL + BUSINESS EVIDENCE

Leakage-Safe Methodology

39.39%

original test rows
overlapped training text

33,042

corrected modeling
rows

6,609

locked final-test
rows

0

normalized-text overlap
groups

- 1

Audit. Detected duplicate-text leakage in the original row-level split.
- 2

Remediate. Removed conflicting-label groups and repeated same-label extras.
- 3

Split safely. Used group-aware development/test boundaries and five fixed development folds.
- 4

Lock before scoring. Selected both routing policies from development evidence only.

Frozen Models and Policies

Item	V1	V2
Model	TF-IDF + Linear SVM	DistilBERT
Top score	0.08	0.22
Top-two margin	0.73	0.91
Artifact size	3.23 MiB	256.35 MiB
Batch throughput*	3,442 rows/s	15 rows/s

*Hardware-specific local benchmark; not a production service-level result. Both score families are uncalibrated model signals.

Retrospective 2025 Comparison

The 30,156-row primary cohort is the headline result; the 49,225-row secondary cohort is the sensitivity view. The 2025 comparison is retrospective and is not a new untouched Version 2 holdout; neither model or policy changed.

Cohort / model	Accuracy	Macro F1	Weighted F1	Coverage	Routed acc.	Misroute
Primary V1	0.8315	0.7527	0.8306	0.7251	0.9203	0.0797
Primary V2	0.8404	0.7620	0.8364	0.7571	0.9174	0.0826
Secondary V1	0.8771	0.7569	0.8766	0.7809	0.9424	0.0576
Secondary V2	0.8751	0.7586	0.8701	0.8007	0.9431	0.0569

Business and Risk Findings

- V2 strengths: higher shared-benchmark Accuracy, Macro F1, and routing coverage.
- V1 strengths: smaller artifact, far faster local inference, slightly better routed-risk results, and completed temporal validation.
- Category risk: V2 primary misroute rates were highest for student loan (32.08%), debt collection (29.84%), and money transfer (24.20%).
- Recommendation: retain V1 as the benchmark and practical low-compute option; keep V2 as the frozen challenger for a new untouched evaluation period.

Limits, Next Steps, and Artifacts

- Sampled public CFPB data is not an institution-specific operational queue.
- Scores are uncalibrated; the 256-token ceiling truncates a material share of transformer inputs.
- Routing limits are project assumptions; no fairness, monitoring, privacy, governance, or production approval is claimed.
- Future promotion requires a new untouched period, category-aware review design, and controlled calibration/monitoring work.

Python

Pandas

NumPy

SciPy

Scikit-learn

PyTorch

Transformers

Jupyter

unittest

Explore the repository

github.com/rihua-tech/financial-complaint-auto-routing-nlp